NIKITA BHANSALI

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EDUCATION

Vellore Institute of Technology — September 2023 – October 2027

Bachelor of Technology in Electronics and Communication Engineering; CGPA:8.48

SKILL SET

Programming Languages: Python, Java
Databases: MySQL, MongoDB
Frameworks: MERN Stack,GitHub

INTERNSHIP

MPOnline (Government of MP) | *AI/ML & Advanced Software Engineering Internship* — May 2026 – July 2026

-Built ML models (regression, classification, decision trees, random forest, SVM, clustering, PCA) using Python, NumPy, Pandas, and Scikit-learn, and developed neural networks with TensorFlow/PyTorch for computer vision and NLP.
-Practiced full-stack SDLC: DBMS, SQL, OOP, Git/GitHub, and CRUD app development, followed by requirement analysis, system design, backend development, testing, and cloud deployment for a capstone project.

PROJECTS

Serenity – AI Therapeutic Companion | *React, TypeScript, Python, WebSockets, Supabase* — July 2026 – September 2026

-Engineered a real-time AI therapist web application capable of analyzing user emotions through continuous facial expression tracking and audio input analysis.
-Developed a full-stack system using React, TypeScript, and Supabase on the frontend, alongside a Python backend integrating Speech Emotion Recognition (SER), ASR, and TTS engines via WebSockets.
-Synchronized multi-modal emotion data (face and voice) to generate context-aware, empathetic responses, resulting in a highly responsive conversational interface for accessible mental health support.

Home Automation with LumaSync | *IoT, Embedded Systems, ESP32* — January 2025 – April 2025

-IoT-based home automation and security system regulating room temperature, detecting motion, and capturing images of entrants, controllable via a local web dashboard and Google Assistant.
-Assembled the system using an ESP32/ESP32-CAM, a DHT11 sensor, a PIR sensor, and a 4-channel relay module; developed a real-time dashboard (HTML/CSS/JS) and integrated Google Assistant via IFTTT.
-Devised predictive, trend-based temperature control logic (dynamic hysteresis) to reduce overshoot and improve system responsiveness.

Smart Surveillance for Public Transport | *Computer Vision, Embedded AI* — September 2024 – January 2025

-Real-time monitoring system for public transport to detect anomalies and enhance passenger safety.
-Implemented OpenCV and YOLO for real-time object detection and video analysis on embedded/edge hardware.
-Fine-tuned anomaly-detection models and optimized YOLO inference for real-time performance on live feeds.

CERTIFICATIONS

Health Hackathon 2026 – Finalist — [tinyurl.com/HealthHack26]

Health Hackathon 2025 – Finalist — [tinyurl.com/HealthHack25]

EXTRACURRICULAR

Languages: Proficient in English, Hindi; learning Japanese and German (300+ day Duolingo streak). [tinyurl.com/300Duolingo]

Leadership: Student Partner at IEEE and VITronix Club, organizing technical workshops for 1000+ students.

National Service Scheme: Served as Hospitality Lead for the NSS Unit Camp and led the Tree Plantation Drive team.

HOBBIES

Anchoring, Dancing, Event Management, Photography, Public Speaking.